Data-Driven Inventory Optimization for an Online Book Retailer

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<u>Executive Summary</u>

Knowing what inventory to stock for an online book retailer is crucial to success. Inventory must be selected strategically, ensuring the right products are chosen to drive sales and customer engagement. Overstocking less demanded products can lead to a waste of space and money, while understocking highly sought after titles inhibits the opportunity for sales. Which best-selling books should an online based retailer prioritize to maximize their revenue and customer engagement? Using data from popular online retailer Amazon allows current consumer book buying trends to be analyzed and recommendations to be made regarding what genres, subgenres, titles and series should be prioritized for this other online retailer to ensure success.

The dataset includes different variables from Amazon. The raw dataset contains 2200+ lines of data, with columns detailing all aspects of a book's metadata - ASIN, title, author, genre, media format, final price, sale price, average rating, number of ratings, etc. Existing variables heavily focused on in this project are Genre, Subgenre, Review Count (used as Customer Engagement), Final Price, Title, Series, and revenue which was calculated by multiplying final price and review count, giving an estimate of how much the title grossed.

After a thorough analysis of the dataset including multiple queries and visualizations, the following are the main key findings presented. Literature & Fiction (referred to hereafter as just Fiction) is an overwhelmingly dominated genre for revenue grossing around 87.7 million USD annually. This is followed by Children's Books (46.8 million), Mystery & Thriller (33.1 million), Teen & Young Adult (18.7 million) and Biographies & Memoirs (14.3 million). Each genre was divided into subgenres, showing an in depth look at high grossing contributors within. Fiction was just broken into Literature and Fiction as subgenres and was omitted from the subgenre comparison due to this lack of further information. Within the Children's Books genre that grossed 46.8 million, Growing Up & Facts of Life subgenre is the highest contributor with 24.1 million. The other four genres, Fiction, Fairy Tales & Myths, Education & Reference and Arts, Music & Photography make up the remaining 22.7 million, which is still less than the one subgenre of Growing Up & Facts of Life. Assigning individual titles and series with a priority score, consisting of the final price and review count, helps compare top performing and titles and how to prioritize them. *Where the Crawdads Sing* was the highest scoring individual title, grossing around 2.4 million USD annually and assigned a score of 10.04, followed closely behind by *Becoming* (2.7 million / 8.66 score) and *A Promised Land* (1.9 million / 6.73 score). Note that the fourth contributing book in this lineup *The Alchemist Gift* grossed 2.4 million but has a lower priority score of 6.51 than its successor; this primely displays how priority score is an impactful metric to display prioritization of titles. The *Harry Potter* series dominated all other series identified in this dataset grossing 17.5 million USD in sales and earning a priority score of 3.92. This was omitted in later analyses to help with clarity. *The Hunger Games* and *Harry Potter* series should be high priority given their high grossing revenue and customer engagement. Series such as *Divergent*, *Fifty Shades of Grey*, and *Atlee Pine* should be given growth opportunities as they have a lower revenue generation but higher customer engagement.

Stocking products in high performing genres and subgenres ensures relevant and sought-after titles are available. Stocking high performing titles such as *Where the Crawdads Sing*, *Becoming* and *A Promised Land* as well as series such as *Harry Potter*, *The Hunger Games,* and *Game of Thrones* all ensure positive revenue grossing and customer engagement. Using metrics such as revenue and customer engagement in tandem can create a solid picture of what a consumer is interested in and what they will pay for it, creating a foundation for the business to base their inventory and prices off of. Assigning the priority scores to products creates the opportunity for scalable inventory opportunities by comparing scores against each other. Stocking not only all titles within high priority scoring series but focusing on the high performing individual titles within a series can even further ensure what the consumer wants is available.

The Amazon dataset gives an inside look into purchasing trends of current consumers, which can be used to positively project inventory purchasing needs for this online retailer. Looking at book metadata and similarities amongst best-selling titles can lead to a comprehensive list of characteristics to look for in future titles. Combining revenue and engagement metrics allows for a full picture look at what is selling well and what the consumers are engaged with, leading to strategic planning for the business. Following these recommendations and ensuring the correct products are stocked will propel this business into success with a loyal consumer base.

Problem Statement

How can an online book retailer identify which titles should be prioritized to maximize revenue and customer engagement? Online book retailers face a myriad of implications, many directly involving inventory. With a retailer operating solely online and not having any brick-and-mortar locations, this opens up the possibility of a much larger catalog of items than a physical location because storage space can be exponentially larger. However, this also poses issues of what makes up that inventory. Customer demands are always changing with the release of new titles, social media trending titles, or other factors. Being competitive with inventory and pricing is what will give a retailer an edge to succeed over their competitors. Having the titles the consumer is actively engaging with and reasonable pricing will lead to not only positive revenue but also customer satisfaction, leading to recurring transactions.

As a stakeholder in this online book retailer, peace of mind that this business will succeed is of utmost importance. Knowing that there are strategic, data-based decisions being made regarding inventory selection ensures minimal waste of time, space and money. Regardless of role in this business, whether that's owner, operations manager, accountant, marketing agent, or something else, the success of all roles depends solely on how the inventory being sold is performing. Without informed decisions being made about inventory items, all roles can be negatively impacted. However, utilizing the data presented and the recommendations provided, this online retailer can successfully operate with positive revenue growth and customer engagement.

Data & Methodology

The dataset utilized in this analysis, 'Amazon_popular_books_dataset.csv'came from a GitHub repository that the user accessed directly from Amazon's semi-publicly published dataset. The GitHub repository can be found at https://github.com/luminati-io/Amazon-popular-books-dataset . The dataset was provided in CSV or JSON format, and contained 2200+ lines of raw, uncleaned data. Types of variables included ASIN, title, author, genre, media format, final price, sale price, ratings count, average rating, and others. See key variables below.

Key Variables

Variable	**Description**
Genre	Primary category
Subgenre	Secondary category; more detailed description of contents
Title	Individual book title
Series	Book series or franchise
Final Price	Retail book price at time of dataset pull
Review Count	Number of customer reviews on Amazon listing
Revenue	Derived metric; final price * reviews
Priority Score	Calculated metric; [(standardized revenue *.06) + (standardized engagement *0.4)]

Cleaning the data took place in both Microsoft Excel Power Query Editor (Version 2604 Build 16.0.19929.20136) and RStudio (Version 2026.04.0 Build 526) using R (Version 4.4.1). An initial examination of the data was done through importing the CSV file into Excel Power Query Editor and transforming through there. The first step taken was verifying the data type assigned to each column was accurate, which all but one were. This inaccurate column was later deleted as the information within was not needed. From here, the columns were reordered to follow a comprehensive format, starting with ASIN, followed by title, author, genre, final price, format, average rating, reviews count, and ending with best seller rank. There were other columns throughout that were not utilized and later deleted, and additionally unnecessary columns were deleted (description, answered questions, upc, etc.) after reorganization. Columns were renamed to align with their use in this dataset, with changed columns including brand to author, format to formats, and categories to genre. The genre column was then split into genre and subgenre at the first delimiter that separated the primary genre from its secondary genre. This new split column was renamed to subgenre and all additional characters were removed. Both the genre and subgenre column were left with just their core classification. The format column was also split into six different columns - one for the format type (Kindle, paperback, audiobook, etc.) and one for format price, for each of the three given formats. These were later renamed to format_type_1 and format_price_1, format_type_2 and format_price_2, and format_type_3 and format_price_3 for querying. The author column was split at the first delimiter and the secondary column was deleted, leaving one core author for each title. At this point, the dataset was mostly clean and ready to move to RStudio.

Within RStudio utilizing an RMarkdown file, the dataset continued to be cleaned with multiple queries and scripts. The main issue when trying to clean this dataset was within the title and author columns. Many titles had the same core title but various editions (Collector's Edition, Special Edition, Hardcover Edition, etc.). This caused numerous lines of data for one core title, and made querying inaccurate. The first solution was to remove all special characters or specific words within a title, however this didn't solve each instance of this. The overall solution for this was assigning each title a core_title and nesting any additional versions underneath. This ensured the data integrity was kept, but allowed queries to process cleanly. At this time, books within a series were also grouped together as families based on their core title and definitions, which aided in querying later on. The author column also had issues with multiple authors still populating instead of just one, and further cleaning was done in R to eliminate these. Additional data cleaning within R included removing any missing values, standardizing the title and series names (all uppercase minus common words), and the creation of the custom revenue variable to display an estimated gross price. At this point, the dataset was clean and transformed into a viable format for querying and analysis to begin.

The creation of the revenue variable was to give a quantifiable metric that could be used to show an estimated gross annual revenue for each title, and was calculated by multiplying final_price by reviews_count. The resulting variable was then utilized in queries and manipulated to reflect each area of querying (ex. Genre_revenue for revenue for each genre and family_revenue for revenue of each series). Customer engagement was consistently measured with reviews_count, and later standardized along with revenue due to their vastly different measurement scales. Both variables were transformed into z-scores using standardization, and allowed for easy comparison. The priority score was calculated by the following formula: Priority Score = (0.6 * standardized revenue) + (0.4 * standardized reviews_count). Revenue was more heavily weighted at 60% due to the business' primary goal of maximizing profits, which revenue directly reflects. Customer engagement represented by reviews_count was weighted with 40% because it's still an important factor in the overall success however revenue is slightly more relevant to overall success for the business. The combination of both the calculated variables and standardization created a solid foundation for further comparative queries to begin.

RStudio was the primary platform used for querying and visualization utilizing a RMarkdown file. Many packages were utilized within R for this project, including ggplot2, dplyr, tidyverse, scales, and patchwork. Once these packages were all installed and loaded into the library, introductory analysis began by exploring different summaries and counts of the dataset variables. Refined queries were then completed, many of which were refined and accompanied by visualizations for the final presentation and analysis. Theme creation was done at this point, creating a cohesive standard for titles, subtitles, captions, labels and visuals to use throughout the project. This included standardizing the text, size, face, spacing, colors, etc. and was titled theme_custom.

Revenue ranking was the first commonly used analytical approach, and is demonstrated within genres and subgenres. This was done by grouping each respective genre or subgenre variable and arranging in descending order by revenue. Visualizations for 'Top 5 Revenue-Generating Genres', 'Top Revenue-Generating Subgenres Within Top 5 Genres', and 'Revenue Contribution of Titles Within High-Opportunity Book Series' were the outcome, displaying a visual descriptor of revenue ranking.

Comparative visualizations with small multiples were created for displaying differences of subgenre contributions within a genre and individual titles within a series. To do this, revenue was calculated for the respective variable, arranged in descending order, and sliced to display only the top five contributors if applicable. To ensure consistency with axes and labels, these small multiple plots were created individually and patched together to one visual with the R package patchwork. Each individual plot was padded to allow only 5 rows each, opting to show blank rows for consistency. Individual plot creation took place first, including adding labels for revenue, altering the x and y axis labels, adding a gradient scale, applying the custom theme, and altering any theme components that required alteration. A secondary plot was created patching all the individual plots together, and followed a similar pattern as above. The resulting visuals were 'Top Revenue-Generating Subgenres Within Top 5 Genres', and 'Revenue Contribution of Titles Within High-Opportunity Book Series'.

Standardization by z-score scaling was done to create the priority score, and this variable was used in 'Top Priority Individual Books', 'Top Priority Book Series', and 'Strategic Prioritization of Book Series'. The two bar charts displaying priority individual and series products started by defining what the book families (aka series) were and grouping the titles into these respective families. From here, revenue values were assigned per family and the priority score was utilized to rank the titles or series in descending order listing only the top ten. Both visualizations followed the same process, adding labels to the bars and axis and ultimately applying the custom theme. The prioritization matrix was created utilizing the same book family definition as above, and arranged the different priority scores for the series in a matrix view with four quadrants. The quadrants were defined by different priority score values and labeled accordingly as High Priority, Grow, Low Priority, and Monetize following a clockwise rotation. Titles, labels and margins were refined and the custom theme was applied for final visual creation.

Key Findings

The following findings summarize the most important trends identified through the analysis and provide insight into inventory optimization opportunities for this retailer.

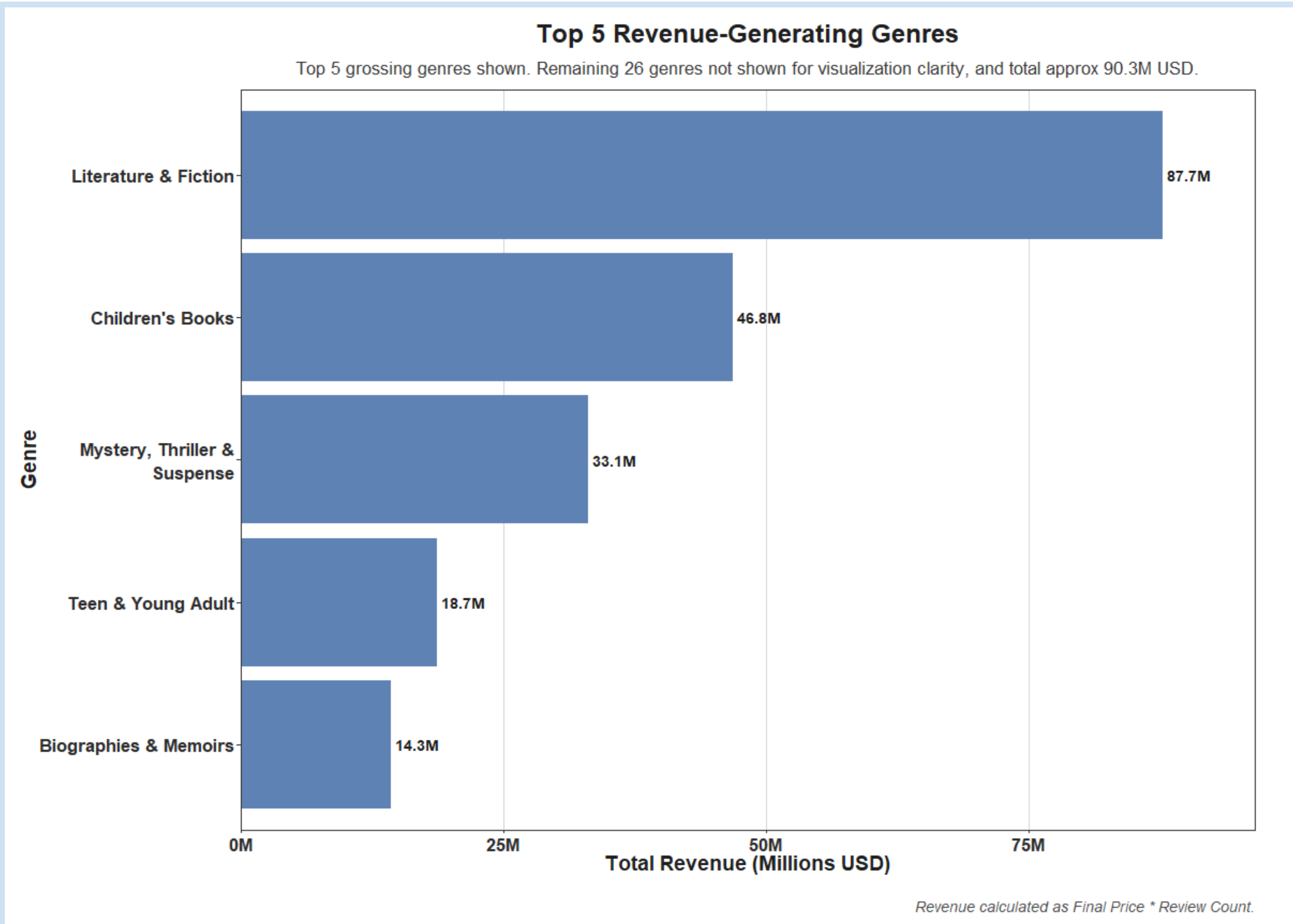

Figure 1

Key finding one from analysis of this dataset is that Literature & Fiction dominates genres with a gross revenue at approximately 87.7 Million USD, as shown in Figure 1. This is followed by Children's Books (46.8 million), Mystery, Thriller & Suspense (33.1 million), Teen & Young Adult (18.7 million), and Biographies & Memoirs (14.3 million). The revenue of these top five genres combined totals almost double what the total revenue of the remaining twenty six genres is, showing these heavily dominate the market. Retailers should heavily prioritize fiction titles, as these grossed the highest and will likely help maximize sales. Genre-level trends help reveal a broad look into customer preferences, and provide solid recommendations for inventory allocation.

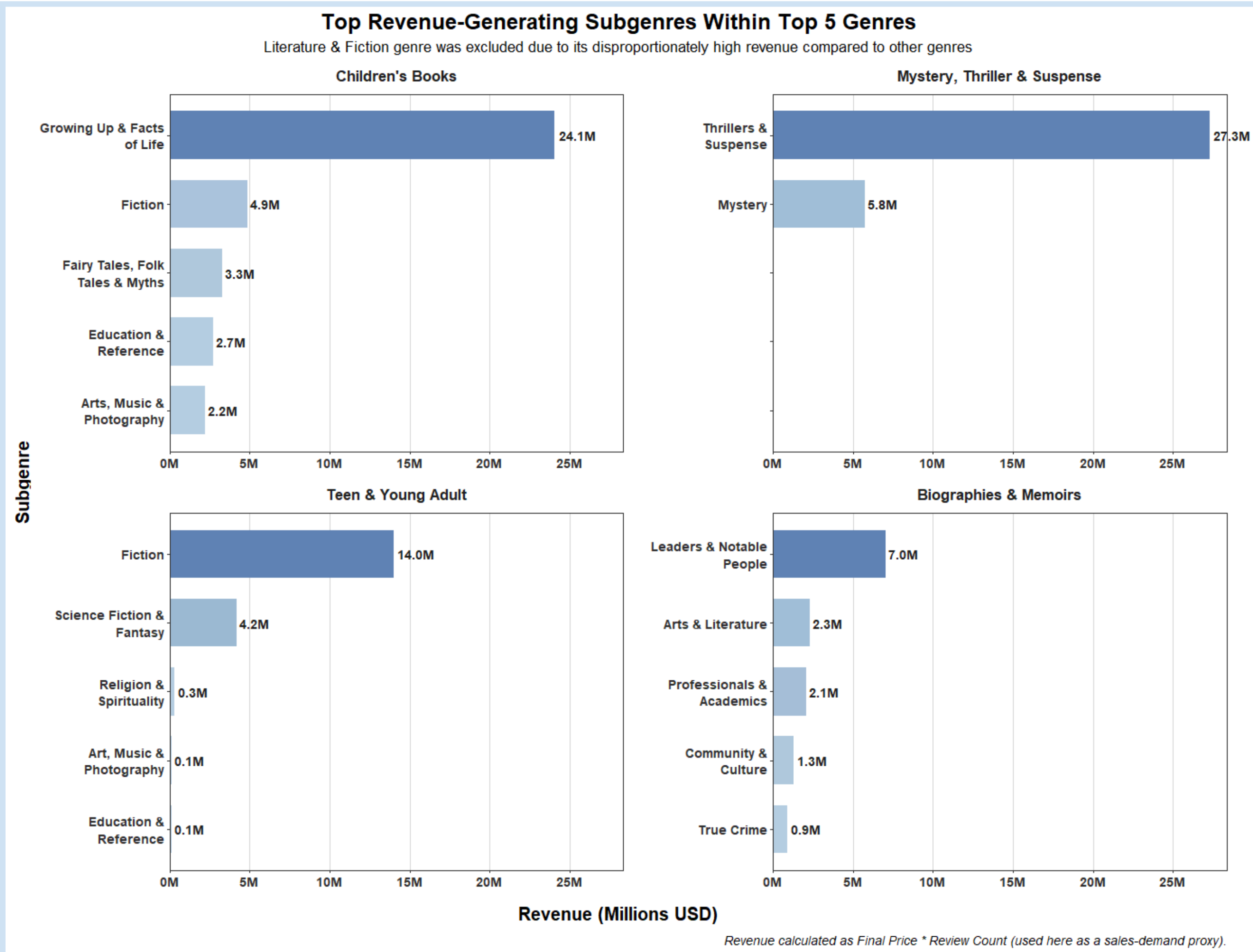

Figure 2

Key finding two within this dataset is that subgenres reveal specific consumer reading preferences and provide a detailed breakdown for inventory selection. The Literature & Fiction genre is omitted from Figure 2 because of its overwhelming domination of the market. Each genre has one revenue anchor that accounts for the majority of overall revenue. Children's Books is dominated by Growing Up & Facts of Life with 24.1 million of the 46.8 million coming from this. Within Mystery, Thriller & Suspense, 27.3 million of the total 33.1 million comes from Thrillers & Suspense titles alone. Teen & Young Adult's total 18.7 million is primarily Fiction with 14.0 million and Science Fiction & Fantasy next in line with 4.2 million. Biographies & Memoirs top contributor is Leaders & Notable People grossing 7.0 million out of the total 14.3 million. These top performing subgenres should heavily be allocated for when stocking inventory as they lead to a catalog meeting consumer demands. Appealing to all subgenres within top performing genres can appeal to a more niche consumer base, drawing in a larger audience in general.

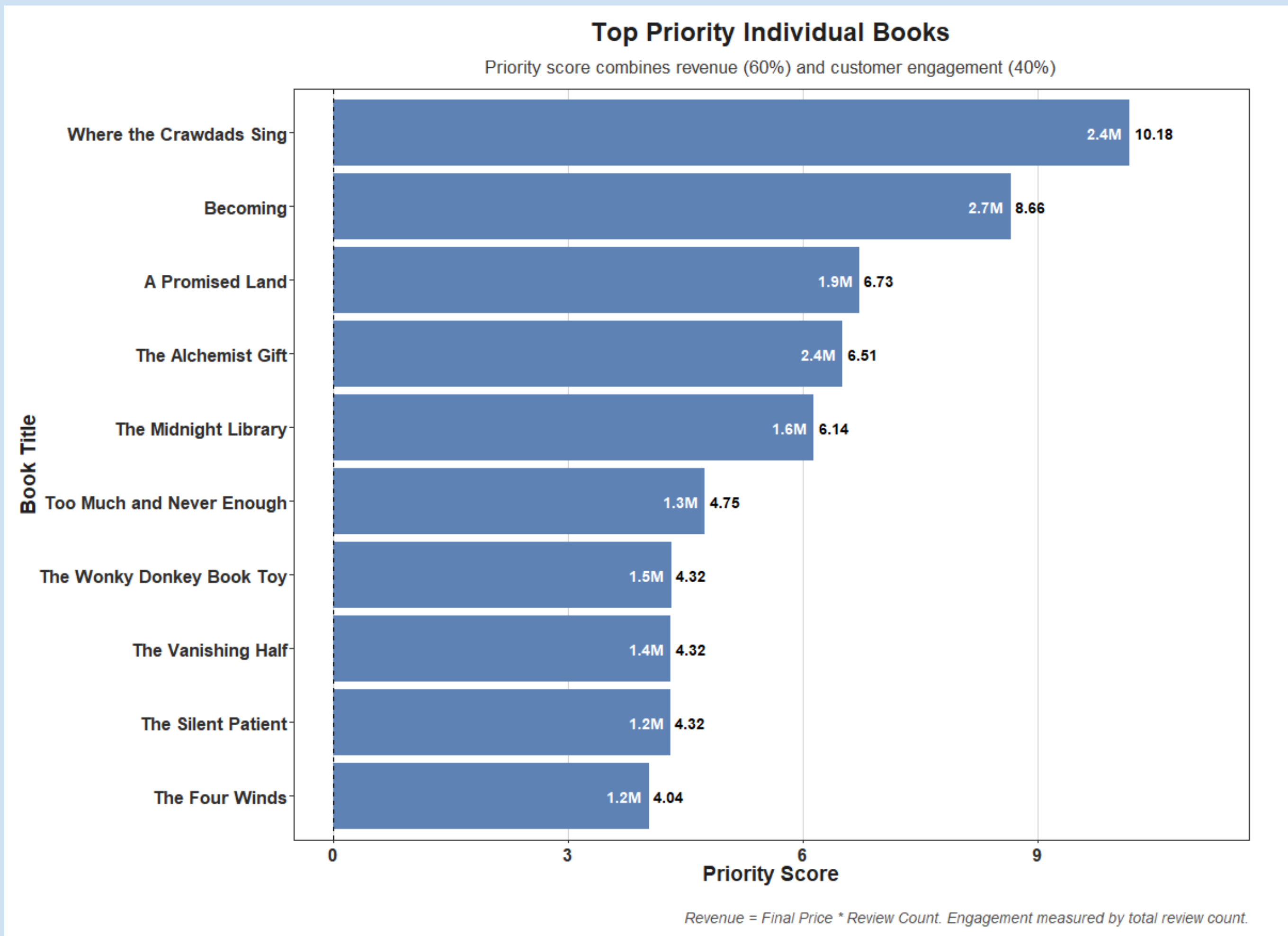

Figure 3

Key finding three details individual top performing titles utilizing a priority score calculated by standardization of revenue and customer engagement, and weighted at 60% and 40% of overall score respectively. These two variables were chosen for this calculation due to their direct correlation and contribution to a successful business. Strong repeat engagement from the consumer suggests a high likelihood of repeat purchases, overall increasing revenue. Identifying and allocating larger numbers of these titles and including them in promotional material constantly engages the consumer while boosting profits.

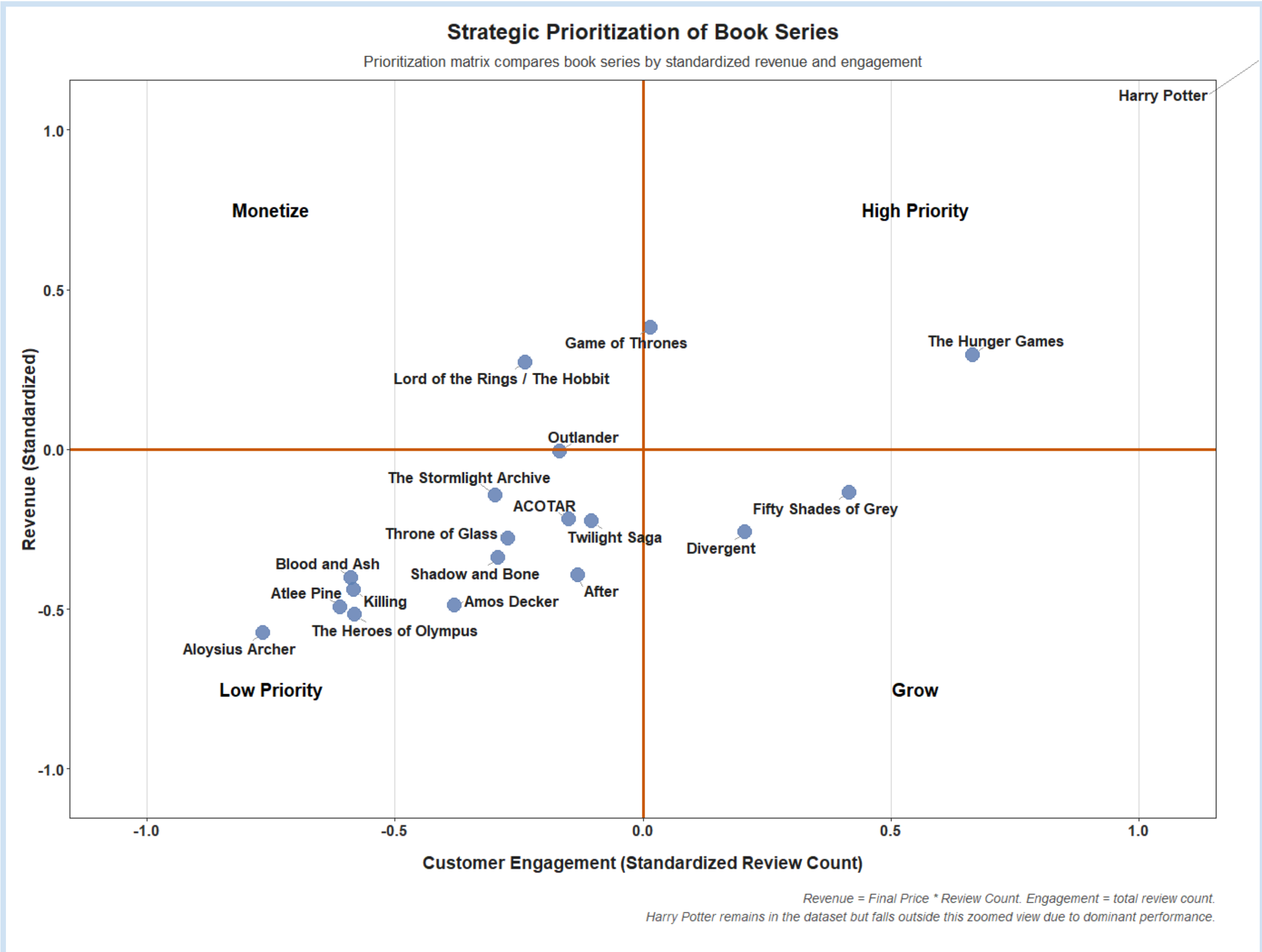

Figure 4

Key finding four is prioritizing inventory of series falling within the High Priority and Grow quadrants of the prioritization matrix shown in Figure 4 to maximize sales opportunities. Series such as *The Hunger Games* and *Harry Potter* fall in the High Priority quadrant due to their strong consumer engagement and revenue, making these products to always keep stocked. Series such as *Divergent* and *Fifty Shades of Grey* in the Grow quadrant demonstrate strong consumer engagement but lower revenue, garnering them the opportunity to grow revenue with promotional materials. *Lord of the Rings / Hobbit* generated strong revenue but had limited consumer engagement, meaning this series should be priced fairly to higher with knowledge a limited amount of consumers will purchase.

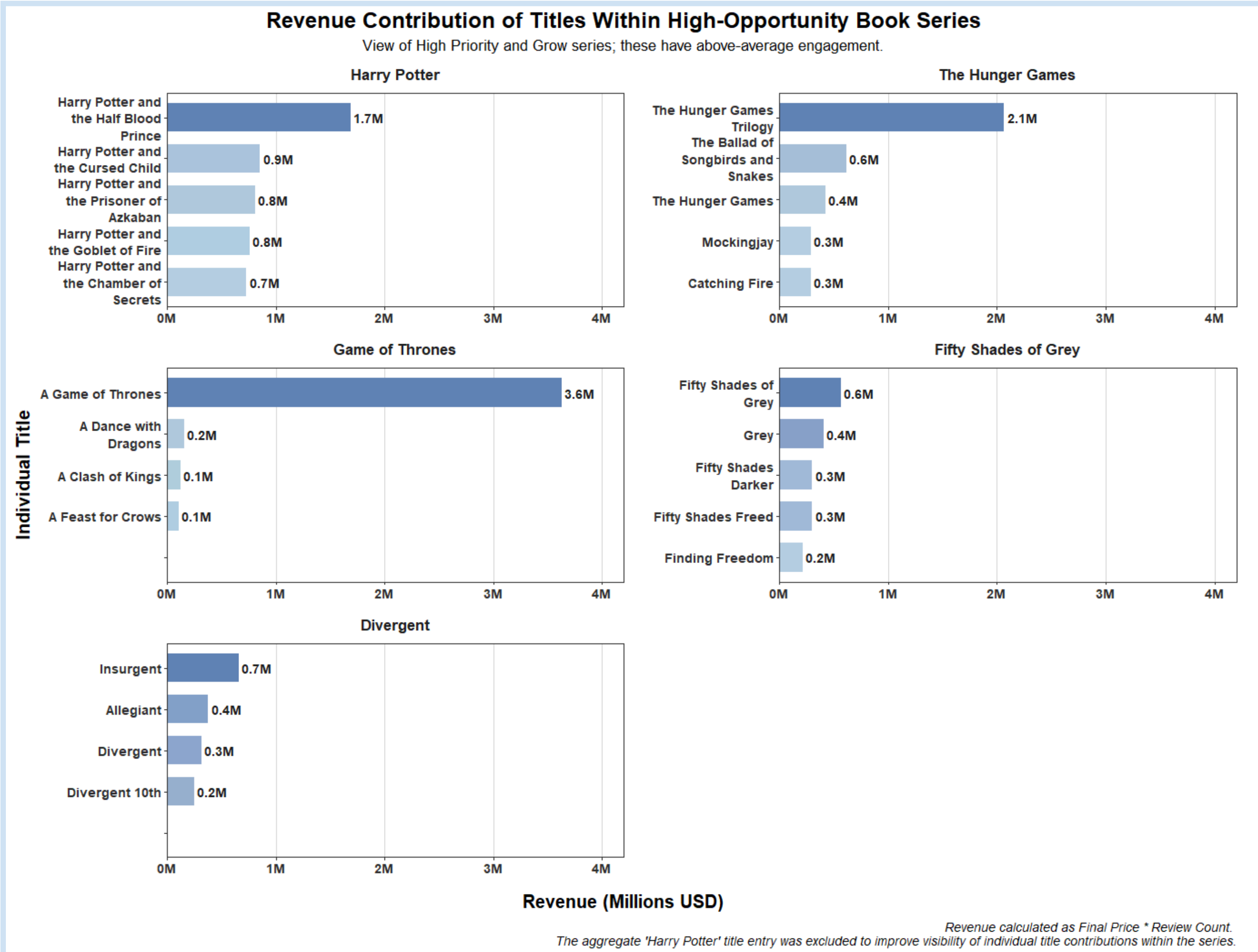

Figure 5

Key finding five is that individual titles within a series can outperform each other and the success of a series does not dictate the success of each title. Many of these series have one title that dominates gross revenue, and the entire collection of titles within a series is commonly the top revenue contributor versus individual titles. Within *The Hunger Games* series, the original trilogy set far outperforms any individual title grossing 2.1 million compared to *The Ballad Of Songbirds and Snakes* grossing 0.6 million, *The Hunger Games* grossing 0.4 million, and *Mockingjay* and *Catching Fire* both grossing 0.3 million. This trend is also demonstrated with the *Game of Thrones* franchise, with the collective box set grossing 3.6 million of the approximately 4 million gross revenue. Consumers tend to gravitate towards newly released titles, entry-point titles, or original trilogies, and these should be prioritized for inventory. This retailer should ensure high performing titles within a major series remain stocked, and monitor for upcoming releases within these series for potential revenue spikes.

Overall, this analysis has demonstrated that the combination of projected revenue and customer engagement creates a stronger foundation for inventory prioritization than relying solely on static sales indicators. Genre, subgenre, title and series-level analyses showed consumer demand tends to concentrate within a small number of high performing products, and retailers should strategically prioritize products within these. Prioritization of these products is likely to lead to maximization of profits and customer satisfaction.

Limitations

This dataset presented some limitations, and therefore it's important to note that the projected revenue and customer engagement metrics are only mathematical estimates. Review_count was utilized as a proxy for engagement and cannot completely represent the actual number of sales of a product. This was deemed a solid representation due to the nature of a highly satisfied or dissatisfied customer leaving a review or rating. Revenue was calculated by final_price and reviews_count and is an estimate of total grossing revenue, and cannot be used in place of actual sales data. The dataset is also limited to only available titles, and didn't account for out-of-stock titles which may have altered data. There was no customer demographic or geographic segmentation, which could have impacted data if there had been. Different retailers than Amazon, different geographical locations and different years will very likely see different results. The dataset was also collected with best-selling titles from 2021, and may have outdated data within. The dataset was deemed appropriate for comparison to this online business despite the above mentioned limitations because it gives an overall analysis of consumer's trends.

Recommendations

This online retailer should prioritize titles within Fiction and other top performing genres. Emphasizing titles in these top grossing subgenres within each top grossing genre ensures relevant titles are available for the consumer, but including titles from all subgenres within a genre can appeal to a more niche consumer base. The retailer should focus on series from the High Priority quadrant from the prioritization matrix and price these similar to the market price. These series have a high customer engagement value as well as overall revenue, and consistently stocking them leaves more opportunity for the consumer to purchase. Monitoring series from the Grow quadrant can provide the business with series to include in the future as they continue to grow. Series within the Monetize quadrant should be priced higher than normal products, as they have an intimate fan base but produce higher revenue. Prioritizing high grossing individual titles within a series and ensuring they are always stocked leads to a higher likelihood the consumer is able to purchase what they are looking for, returning more in revenue. However, ensuring all titles within a high priority series increases chances of multiple books being purchased at once. Running franchise based promotions would encourage consumers to make multiple product purchases, overall driving both consumer engagement and revenue.

For the business to have tangible data of their own for future analysis, I recommend tracking overall sales amount, repeat customer transactions, and seasonal demand patterns. Keeping track of overall sales amount on a daily, weekly, monthly, quarterly and yearly basis helps project future sales patterns and inventory selections. Recording repeat customer transactions is a solid metric to measure customer retention and engagement, and see how consumers are feeling about the retailer. Repeat customers usually means positive interactions are happening, whereas a low number of returning customers could lead to the opportunity for improvements. Seasonal demand patterns can aid in inventory planning, especially if there seems to be a season where consumers are more or less active. Ensuring more inventory is on hand during these spikes enforces more sales to be made, and stocking less inventory during the slower seasons can allow for funds to be utilized elsewhere.

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